

Data Collection and Preprocessing Phase

Date	4 october 2026
Team ID	xxxxxx
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains agricultural crop production information from India. The data includes fields related to State, Crop, Year, Season, and Production. The dataset was maintained in Excel format and used as the primary data source for Tableau. The data was explored to understand its structure, fields, values, and suitability for agricultural production analysis.
Data Cleaning	The dataset was checked for missing values, duplicate records, inconsistent values, and incorrect entries. The available data was reviewed and retained where it was suitable for analysis. Unnecessary fields or records, if any, were excluded before visualization.
Data Transformation	The dataset was organized and prepared for Tableau analysis. Required fields were arranged according to their purpose as dimensions and measures. Data was filtered and sorted where required to create state-wise, crop-wise, year-wise, and season-wise visualizations. Calculated values and aggregations were used in Tableau to support KPI cards and production analysis.
Data Type Conversion	The data types of the required fields were reviewed and corrected where necessary before analysis. Year was treated as a date/year-related field, while Production was treated as a numerical measure. State, Crop, and Season were treated as

	categorical fields for filtering and visualization.
Column Splitting and Merging	No major column splitting or merging was required for the agricultural production dataset. The existing State, Crop, Year, Season, and Production fields were used directly for analysis and visualization in Tableau.
Data Modeling	The project uses a single agricultural production dataset, so no relationships between multiple tables were required. The required dimensions and measures were organized within Tableau for analysis. State, Crop, Year, and Season were used as dimensions or filters, while Production was used as the primary measure for visualization and KPI analysis.
Save Processed Data	The prepared agricultural dataset was maintained in Excel format and connected to Tableau for visualization. The Tableau workbook containing the prepared data, visualizations, dashboard, and Tableau Story was saved for future use and project submission.